

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III B. TECH II SEMESTER

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Batch Id : 23MPCS-H21

Title of the Project : Real-Time Vehicle Detection and Traffic Flow Prediction

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Batch Id : 23MPCS-H21

Title of the Project : Real-Time Vehicle Detection and Traffic Flow Prediction

Objectives

1. To develop an intelligent traffic control system using image processing and real-time vehicle detection.
2. To analyse traffic density at a four-way junction and dynamically adjust signal timings based on vehicle count.
3. To detect and prioritize emergency vehicles (like ambulances, fire trucks, and police cars) for uninterrupted passage.

Outcomes

1. Dynamic Traffic Control: Successfully implemented a system that adjusts signal timings automatically based on real-time traffic density.
2. Emergency Vehicle Priority: Enabled automatic detection and prioritization of emergency vehicles, reducing their response time.
3. Reduced Congestion: Achieved smoother traffic flow and minimized vehicle waiting time at intersections.

ABSTRACT

The escalating complexity of urban traffic networks has underscored the limitations of conventional fixed-time signal controllers, which remain insufficiently responsive to dynamic spatiotemporal variations and often hinder the unobstructed movement of emergency vehicles. This study presents an intelligent, adaptive traffic signal regulation framework that employs a real-time processing pipeline aligned with the conceptual flow diagram, wherein heterogeneous traffic inputs are captured, computationally interpreted, and translated into optimized signal phase decisions. Utilizing YOLO-based deep learning image analytics, the system continuously estimates vehicular density at a four-leg intersection and performs high-precision identification of emergency vehicles under varying environmental and traffic conditions. Experimental observations demonstrate notable improvements in intersection throughput, reduced queue lengths, and enhanced temporal efficiency during peak-load scenarios. Due to its modular design and compatibility with existing traffic infrastructure, the proposed framework exhibits substantial feasibility for practical, large-scale implementation in modern urban environments. Future enhancements may include coordinated multi-junction optimization, integration of predictive traffic modeling using historical and real-time datasets, and deployment of communication interfaces enabling interaction with connected and autonomous vehicles. Collectively, these advancements have the potential to further elevate the system's effectiveness, establishing a robust and scalable solution for next-generation urban traffic management.

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